

SEMESTER EXAMINATION (DEC 2023)
CLASS: B. Tech SEMESTER: I
PAPER CODE: BEM-C102
PAPER TITLE: ENGINEERING MATHEMATICS-I

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: (i) Expand $e^{\sin x}$ by Maclaurin's series upto the terms containing x^4 .

(ii) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$

Question-2: (i) Show that radius of curvature at any point of the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ is $4a \cos \theta$

(ii) Find the asymptotes of the curve $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$

Question-3: (i) If $u = \log \frac{x^4 + y^4}{x + y}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

(ii) Find the equations of the tangent plane and normal to the surface $z^2 = 4(1 + x^2 + y^2)$ at (2, 2, 6).

Question-4: If $u = xyz$, $v = x^2 + y^2 + z^2$, $w = x + y + z$, find $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.

Question-5: Show that area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $16a^2/3$.

Question-6: Prove that $B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m + n)}$

Question-7: Prove that $r^n \vec{r}$ is an irrotational vector for any value of n but is solenoidal only if $n+3=0$

Question-8: Verify Green's theorem in the plane for $\oint_C (xy + y^2) dx + x^2 dy$ where C is closed curve of the region bounded by $y = x$ and $y = x^2$.

Question-9: Define normal form of a matrix. Reduce the given matrix into normal form and hence find its rank.

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Question-10: Investigate the values of λ and μ so that the equations

$$2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$$

have (i) No solution (ii) a unique solution (iii) an infinite number of solutions

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

Question-1: If $x = \sin\left(\frac{1}{a} \log y\right)$ then prove that $(1 - x^2)y_{n+2} - x(2n + 1)y_{n+1} - (n^2 + a^2)y_n = 0$ and hence using McLaurin theorem show that $e^{a \sin^{-1} x} = 1 + ax + \frac{a^2 x^2}{2!} + \frac{a^2(1+a^2)x^3}{3!} + \dots$

Question-2: (i) In a plane triangle, find the maximum value of the $\cos A \cos B \cos C$

(ii) If $u = f(r)$ and $x = r \cos \theta, y = r \sin \theta$ prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

Question-3: (i) Find the area enclosed by the curve $a^2 x^2 = y^3(2a - y)$.

(ii) Find the volume of the solid generated by revolving the lamina $r^2 = a^2 \cos 2\theta$ about the line $\theta = \frac{\pi}{2}$.

Question-4: (i) Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\mathbf{i} - (2xy)\mathbf{j}$ taken around the rectangle bounded by $x = \pm a, y = 0, y = b$.

(ii) Apply Gauss's Divergence theorem to evaluate $\iiint_S [(x^3 - yz)dydz - 2x^2 ydzdx + zdx dy]$ over the surface of a cube bounded by the coordinate planes and planes $x = y = z = a$.

Question-5: Verify Cayley Hamilton theorem for matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find its inverse.

Question-6: Expand $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$ in power of $(x-1)$ and $(y-1)$ upto third terms. Hence compute $f(1.1, 0.9)$ approximately.

Question-7: Find the Volume of the Greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Question-8: (i) Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

(ii) Prove that $\beta(m+1, n) + \beta(m, n+1) = \beta(m, n)$.